Shiyu Yue

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EDUCATION

The Chinese University of Hong Kong, Shatin, N.T., Hong Kong
Mphil in Physics, Faculty of Science — Aug. 2024 – Present

Sun Yat-sen University, Guangdong, P.R.China
Bachelor in Physics, School of Physics and Astronomy — Sept. 2020 – July. 2024

PUBLICATIONS

Yue, S., Tang, H. et al. (2025). Can I trust you?: Interpreting radio galaxy classifier with FR-DEEP dataset and eXplainable AI.” ApJS, Under revision.

Yue, S., Feng, L. et al. (2024). Pair counting without binning–a new approach to correlation functions in clustering statistics. MNRAS, 535(4), 3500-3516.

Tang, H., **Yue, S.** et al. (2023). A model local interpretation routine for deep learning based radio galaxy classification. In 2023 XXXVth General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS) (pp. 1-4). IEEE. (co-first author)

RESEARCH EXPERIENCE

How to Fit All Emission Lines Simultaneously with Photoionisation Models

Aug. 2024 - Present

Interstellar Medium
Advisor: Prof. Renbin Yan, The Chinese University of Hong Kong ; Mphil Thesis

- Developed a Python pipeline for MaNGA integral-field spectroscopy. Extracted and calibrated datacubes, measured optical strong-line ratios, and mapped star-forming regions into 3D reprojected BPT space for direct comparison with photoionisation models.
- Built and parallelised a Cloudy-based photoionisation framework. Generated large grids across gas and stellar metallicity, O–N–S abundance patterns, ionisation parameter, and SED–metallicity mapping. Used 3D line-ratio reprojections and joint Bayesian inference to rank models against MaNGA observations.
- Applied the framework to MaNGA star-forming regions. Showed that standard models over-predict [S III] λ9530 by a factor of three. Identified revised models that halve the [S III]/[S II] discrepancy while preserving good fits to [N II]/Hα, [S II]/Hα, and [O III]/Hβ, improving strong-line calibrations.

A Fast Statistical Algorithm for Cosmology Based on Multiscale Analysis and Application

Oct. 2021 - Oct. 2024

Cosmic Statistics
Advisor: Prof. Longlong Feng, Sun Yet-sen University ; Bachelor Thesis

- Developed a novel theoretical framework for correlation functions by re-conceptualizing binned pair-counting as an 'in-situ' convolution of the density field, enabling a highly efficient **O(N log N)** algorithm that directly incorporates binning effects into theoretical models.
- Applied the algorithm to compute the three-point correlation function (**3PCF**) for $\sim 10^8$ particles **MDPL2** Dark Matter Simulation, achieving $\leq$5% deviation from binning-corrected tree-level perturbation theory at scales $r > 20\ h^{-1}$ Mpc. The analysis can be completed in under 8 hours with GPU acceleration (NVIDIA A800).

- Developed a theoretical framework for the 3PCF incorporating binning effects via filtered density fields, enabling high-precision binning 3PCF modeling.
- Optimized filter radii (3-8h^{-1} Mpc) and sampling density to minimize shot noise in **Quijote** halo catalogs, quantified trade-offs between binning bias and cosmic variance to establish adaptive filtering as optimal for sparse tracers (n $\sim 10^{-4} h^3 Mpc^{-3}$).

Explainable Artificial Intelligence Applications on Radio Galaxy Morphology Classification
Jul. 2021 - Aug. 2021 / Sept. 2022 - Dec. 2024
Radio Astronomy & Machine Learning
Advisor: Dr. Hongming Tang; Xi'an Jiaotong-Liverpool University

- Designed and implemented a Convolutional Neural Network (**CNN**) for radio galaxy morphology classification.
- Engineered a custom interpretation pipeline leveraging Local Interpretable Model-agnostic Explanations (**LIME**), integrating advanced image segmentation and tailored visualization tools to map model attention to astrophysical features (jets, lobes, hotspots), thereby enhancing transparent assessment of deep learning models.
- Pioneered quantitative analysis of model decision-making with explainable AI, statistically validating the extraction of physically meaningful features and analyzing its misclassification rationale, revealing limitations with less extended sources.

Hunting Central Star of Round Galatic Planetary Nebula
Sept. 2023 - Dec.2023 / Sept. 2025 - Present
Planetary Nebula & Multimodal AI
Advisor: Prof. Quentin Parker; Hong Kong University

- Systematically identified Central Stars of Planetary Nebulae (**CSPNe**) by cross-matching the HASH planetary nebula database with multi-wavelength imaging surveys (PanSTARRS, DECaPS DR2).
- Employed high-precision astrometry to statistically quantify CSPN-nebula offsets, and conducted robust cross-validation against Gaia DR3.
- *Ongoing Work*: Integrating IPHAS, VPHAS+ multiband photometric data and Gaia stellar parameters to develop a multimodal deep learning framework for automated CSPNe classification and candidate selection.

SKILLS

Computer Skills :
- Proficient in Python and C++.
- Proficient in Cloudy Simulation.
- Familiar with Machine Learning (Deep Learning Framework: PyTorch).
- Familiar with Linux, using linux desktop and system daily.

TEACHING EXPERIENCE

Teaching Assistant, The Chinese University of Hong Kong.
- An Introduction of Astronomy — Jan. 2025 - Present
- An Introduction of Meteorology — Sept. 2024 - Dec. 2024

ADWARDS

Postgraduate Studentships, The Chinese University of Hong Kong	2024-2026
Outstanding Graduate, Sun Yet-sen University	Jun. 2024
Outstanding Graduation Thesis, Sun Yet-sen University	Jun. 2024
First award of scholarship, Sun Yet-sen University, three consecutive years	2021-2023
China Aerospace Foundation-Uniasia Aerospace scholarship, Sun Yet-sen University	Nov. 2021